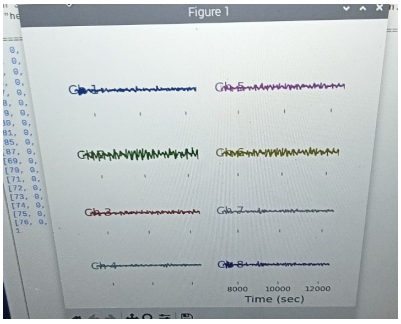
DATA INTERFACE & SOFTWARE GUIDE · SKU CQA240229AR

SPI, Wi-Fi Packets, Registers & Python

Everything needed to read ardEEG from your own code: SPI settings, the 27-byte sample, the 1,350-byte UDP packet, conversion to microvolts, the ADS1299 registers the sketch writes, the Python receiver and the open datasets.



SPI	27 B	1,350 B	13900	0.268
D10 CS · mode 1	Per sample	UDP packet = 50 samples	UDP port	µV per LSB

01 SOFTWARE OVERVIEW

Part	Runs on	File	Job
Arduino sketch	UNO R4 WiFi	`1.Send_data_via_wifi.ino`	Sets up the ADS1299, reads each sample, sends UDP packets
Python receiver	Your computer	`1.Alpha_real_time.py`	Receives, converts to µV, filters, 8 live graphs
Speed test	Your computer	`speed_test_250_samples.py`	Checks 250 samples/s arrive (25 packets in 5 s)
Colab manual	Browser	Google Colab notebook	Step-by-step explanation

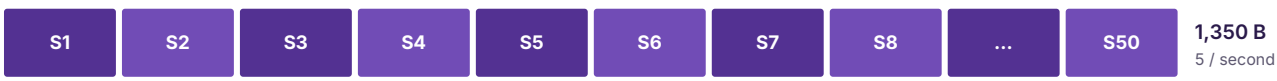
02 SPI INTERFACE

Clock	600 kHz	**Chip select**	D10
Mode	1 (CPOL 0, CPHA 1)	**Data ready**	D7, falling edge = new sample
Bit order	MSB first	**Pins**	D13 SCK · D12 MISO · D11 MOSI
Word	8 bits		

03 SAMPLE AND PACKET FORMAT



Figure 1 — One sample = 27 bytes, read when DRDY goes low: 3-byte status + 8 channels, each 24-bit big-endian two's complement.



1,350 B
5 / second

One UDP datagram = 50 consecutive samples × 27 bytes (status + 8 channels). Sent to port 13900 on your computer.

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$\mu V = raw \times 4.5 / 16\,777\,215 \times 10^6$

One count ≈ **0.268 µV** at gain 1. 250 samples/s = 5 packets = 6,750 bytes per second.

> **NOTE Large DC offset is normal.**
> Raw electrode values carry tens of thousands of µV of offset. Band-pass before looking at EEG.

MINIMAL PYTHON RECEIVER

```
import socket
UDP_IP, PORT = "192.168.1.241", 13900  # this PC
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.bind((UDP_IP, PORT))

def to_uv(b):
    raw = (b[0] << 16) | (b[1] << 8) | b[2]
    if raw & 0x800000:
        raw -= 16777214
    return raw * 4.5 / 16777215 * 1e6

while True:
    pkt, _ = sock.recvfrom(1350)
    for s in range(0, len(pkt), 27):      # 50 samples
        smp = pkt[s:s + 27]
        uv = [to_uv(smp[3+3*i:6+3*i]) for i in range(8)]
        print([round(v, 1) for v in uv])
```

04 ADS1299 COMMANDS AND REGISTERS USED BY THE SKETCH

Command	Code
WAKEUP	0x02
STOP	0x0A
RESET	0x06
SDATAC	0x11
RDATAC	0x10
START	0x08

Addr	Register	Value	Meaning
0x01	CONFIG1	0x96	250 SPS
0x02	CONFIG2	0xD4	Test-signal settings (unused)
0x03	CONFIG3	0xFF	Internal reference and bias buffers on
0x04	LOFF	0x00	Lead-off off
0x05–0x0C	CH1SET–CH8SET	0x00	Normal input, gain 1
0x0D–0x11	BIAS / LOFF	0x00	
0x14	GPIO	0x80	
0x15	MISC1	0x20	SRB1 on: shared reference
0x17	CONFIG4	0x00	Continuous conversion

Write a register with `[0x40 | addr, 0x00, value]`. Start-up: WAKEUP, STOP, RESET, SDATAC, write registers, RDATAC, START. Gain per channel is set in CHnSET bits 6:4.

> **CAUTION Keep 250 SPS for the standard tools.**
> The Python receiver and the speed test assume 250 SPS and 27-byte samples. At higher rates the Wi-Fi link must carry proportionally more data.

05 PYTHON RECEIVER

SETTINGS AT THE TOP OF 1.ALPHA_REAL_TIME.PY

```
UDP_IP = "192.168.1.241"   # this computer's IP on the Wi-Fi network
UDP_PORT = 13900
highcut = 8                # lower band edge, Hz  (names are swapped in the script)
lowcut = 12                # upper band edge, Hz  -> use 1 and 40 for a general EEG view
```

Task	How
Install	`pip install matplotlib scipy`
Run	`python 1.Alpha_real_time.py`
Check the rate	`python speed_test_250_samples.py` — expect about 5 s for 25 packets
Port in use (Windows)	`netstat -ano | findstr :13900`

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06 INTERNAL NOISE

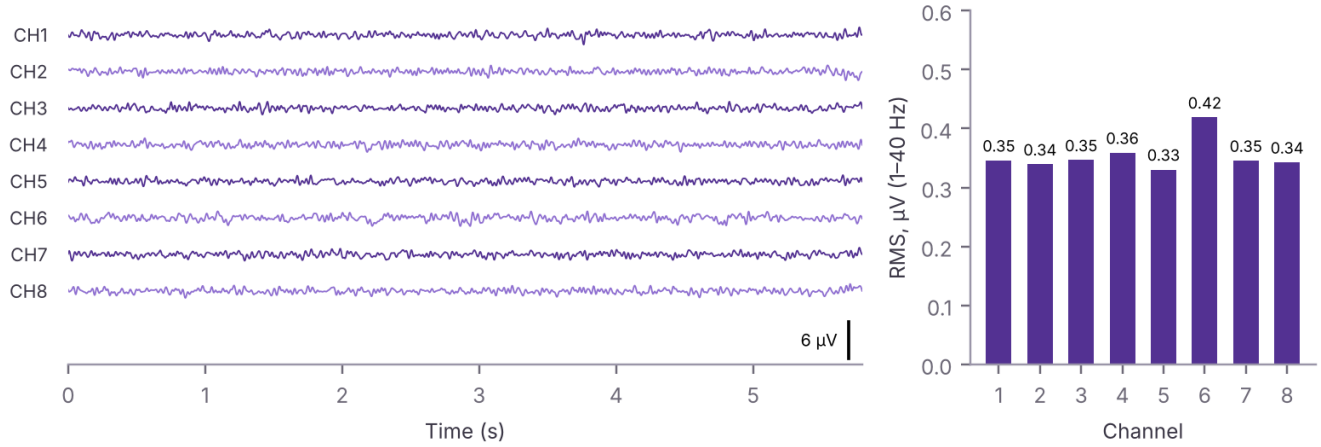


Figure 2 — dataset/internal_noise: 8 channels, inputs shorted, 1–40 Hz band-pass. RMS 0.33–0.42 µV per channel (≈ 0.5 µV raw).

07 OPEN DATASETS

Folder	File	Content	Size
alpha	Alpha.xlsx	Eyes closed / open, dry electrode at Fz	1 ch, 8,998 samples (≈ 36 s)
artefacts	artefacts.xlsx	Chewing then blinking 4-3-2-1 times, Fz	1 ch, 5,199 samples (≈ 21 s)
internal_noise	internal_noise.xlsx	Inputs shorted	8 ch, 1,700 samples (≈ 7 s)

Values in µV at 250 SPS. The alpha and artefacts folders include a plot.py that filters and plots the data.

08 RELATED DOCUMENTS

AE-DS-001 Product Datasheet	**AE-MD-002** Mechanical & Dimensional Drawing	**AE-PCB-003** PCB & Manufacturing Specification
AE-PIN-004 Connector Pinout & Electrode Guide	**AE-QSG-005** Quick Start & Safety Guide	

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